British Journal of Oral and Maxillofacial Surgery (1986) **24**, 323–331
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OSTEORADIONECROSIS: A STUDY OF THE INCIDENCE IN THE NORTH WEST OF ENGLAND

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Summary. A retrospective survey of 200 patients was carried out to determine the incidence of osteoradionecrosis in the North West of England. The patients all had been treated with radiotherapy for squamous cell carcinoma of the oral cavity. An overall incidence of osteoradionecrosis of 19.6% was found. Cases were designated as having major, moderate or minor osteoradionecrosis. The incidence of major osteoradionecrosis was 8%. Hemimandibulectomy was required for management of the necrosis in 3%. Factors involved in the aetiology are discussed.

Introduction

Osteoradionecrosis (ORN) of the jaws is one of the more unpleasant, and, potentially serious side effects, of radiotherapy in the management of neoplastic lesions arising in and around the oral cavity. It can vary from small areas of exposed, non-vital bone with overlying soft tissue necrosis, which heal slowly over many months, to a rapid destructive process with gross bone loss. It can produce a devastating functional and cosmetic deformity. ORN of the jaws was originally noted by Regaud in 1922 and the pathology further described by Ewing (1926a) under the name 'radiation osteitis'. More recently the general pathology has been described by several authors (Rubin & Casarett, 1968; Fajardo, 1978). In the jaws, MacLennan (1955) felt that the effects of radiation on the vasculature were significant in the development of ORN. Ewing (1926a & b) drew attention to structural changes in bone. Watson & Scarborough (1938) pointed out the effect of radiation on the periosteum, but Gowgiel (1960), in his series of experiments aimed at elucidating the pathogenesis of ORN, showed that ORN resulted primarily from the direct effect of irradiation on osteocytes and suggested that the changes seen in the periosteum were secondary to the development of inflammation or necrosis. The changes that occur in the vasculature were responsible for continuing and prolonging the radiation effects. In the jaws, it is noticeable that whereas ORN of the mandible occurs with some frequency, ORN involving the maxilla is rare. Meyer (1958) felt that the mandible is more frequently affected because it is more often in the field of radiation. Experimentally, Gowgiel (1960) showed that when a whole skull was irradiated the mandible was still more frequently affected by ORN and he suggested that as the mandible consists of more compact bone there is greater scattering of radiation with consequently increased radiation to mandibular osteocytes. It has also been suggested that the better blood supply to the maxilla protects it from ORN (MacLennan, 1955; Rubin & Doku, 1976). It seems likely that a combination of these factors is responsible for the noted prevalence in the mandible.

Traditionally, it has been thought that radiation damage was followed by trauma and sepsis to produce ORN (Watson & Scarborough, 1938). The classic scenario was that of direct bone trauma such as caused by the extraction of teeth; however,

(*Received* 28 September 1985; *accepted* 9 October 1985)

ORN has been reported occurring in edentulous jaws by several authors (Beumer *et al.*, 1972; Murray *et al.*, 1980a; Morrish *et al.*, 1981), although some of the cases may have had extractions closely related to radiotherapy (Beumer, 1979).

Gowgiel (1960) found ORN occurring under intact gingivae and stated that such a dormant ORN required disruption of the gingival surface to become grossly evident. Such disruption could occur as a result of spontaneous soft tissue necrosis or be related to teeth, dentures or even rough food. It appears therefore, that whereas trauma is not directly responsible for ORN, it is in many cases responsible for making it clinically evident.

The role of sepsis is also in doubt. Most clinicians will have seen cases of gross necrosis with no evidence of infection. In 1958 Meyer reported that there was suppuration at some time in all of his 26 cases. Guttenberg (1974) felt that ORN should be divided into septic and aseptic types. Marx (1983) felt that microorganisms only played a contaminant role in the aetiology of ORN. Many of the gross cases of ORN described, however, appear to have been infected (Rankow & Weissman, 1971; Saunders *et al.*, 1978; Parulekar & Paonessa, 1980; Dolezal *et al.*, 1982). Marx (1983) has pointed out that radiated tissues are hypoxic, hypovascular and hypocellular. They are unable to cope with normal cellular renewal and hence the development of a chronic non-healing radionecrotic wound. Infection occurring in this situation inevitably leads to further deterioration.

ORN must now be considered to occur as a direct result of radiation damage to the underlying bone. Minor trauma or soft tissue necrosis leads to exposed bone and, in already damaged tissues healing fails to occur and infection may supervene. The incidence of ORN varies considerably from survey to survey (Table I); several factors are thought to influence it. With the older orthovoltage therapy techniques, the energy absorption of bone was approximately two and a half times greater than the surrounding soft tissues. The energy deposited in bone with megavoltage sources is much reduced (Meyer, 1962). With the arrival of megavoltage therapy however, not all studies report a decrease in the incidence of ORN (Coffin, 1964; Rankow & Weissman, 1971). Ewing (1926a) mentions the concept of fractioning the dose to reduce damage. Certainly the total dose given affects the incidence of ORN (Meyer, 1958; Grant & Fletcher, 1966; Rankow & Weissman, 1971;

Table I
The results of published ORN incidence studies

Year	Authors	Site	Dose	No. of Pts.	ORN	%	Years studied
1938	Watson & Scarborough	1,2,4,		1819	235	12.9	1930–1937
1958	Meyer	1,4,	4000–18 000r	491	26	5.3	1940–
1962	MacComb	1,		251	93	37.0	1952–1959
1966	Grant & Fletcher	2	6000–13 000r	176	66	37.5	1954–1962
1971	Rankow & Weissman	1, (2)	5000+r	176	12	6.8	1965–1968
1972	Beumer *et al.*	1, (2)	5000–7000r	278	10	3.6	1961–1969
1972	Daly *et al.*	1,2,		304	66	21.7	1966–1971
1976	Bedwinek *et al.*	1,2,	5000–7000r	381	54	14.2	1966–1971
1980a	Murray *et al.*	1,2,	2000–8000r	404	77	19.0	1971–1975
1981	Morrish	1,2,	5000–7500r	100	22	22.0	1971–1977
1983	Coffin	1,2,3,		2853	22	0.7	1970–1981

Sites: 1=Intraoral 2=Pharynx and Tonsil 3=Salivary Gland 4=Lip

Bedwinek *et al.*, 1976; Murray *et al.*, 1980a; Morrish *et al.*, 1981), but time-dose fractionation characteristics play a role in the development of the acute and intermediate effects of radiotherapy (Bloomer & Hellman, 1975). In some surveys the use of interstitial implants led to a higher incidence of ORN (Beumer *et al.*, 1972; Murray *et al.*, 1980a).

It is generally accepted that in patients with tumours sited close to bone such as floor of mouth and alveolar tumours the incidence of ORN is high (Watson & Scarborough, 1938; MacComb, 1962; Bedwinek *et al.*, 1976; Murray *et al.*, 1980a). In some surveys the tumour size (T stage) is also a factor influencing the incidence of ORN (MacComb, 1962; Grant & Fletcher, 1966; Rankow & Weissman, 1971). Others, however, feel that size has no effect (Murray *et al.*, 1980a; Morrish *et al.*, 1981). Surgery in the irradiated field is another factor that has been shown to increase the incidence of ORN (Rankow & Weissman, 1971; Daley *et al.*, 1972).

Murray (1980b) has shown that there is a relationship between the presence of dental disease in dentate patients, and the incidence of ORN, and many surveys have noted an increased incidence of ORN in dentate patients (Beumer *et al.*, 1972; Murray *et al.*, 1980a; Morrish *et al.*, 1981). The incidence falls when a programme of dental care is instituted (Daley *et al.*, 1972; Murray *et al.*, 1980a). The role of extractions as an aetiological factor in the development of ORN is not in doubt but there is still uncertainty as to the optimum timing of the extractions and indeed to the techniques of extraction. Pre-irradiation dental clearance was favoured by Watson and Scarborough (1938) in the pre-antibiotic era. More recently Rankow and Weissman (1971) suggest dental clearance 10–14 days prior to therapy, the extractions being accompanied by alveoloplasty. Meyer (1958) advocates pre-irradiation extractions plus alveolectomy and feels it is a reason for his fairly low incidence of ORN (5.3%). Beumer *et al.* (1972), Daley *et al.* (1972), and Bedwinek *et al.* (1976) all showed a higher incidence of ORN with pre-irradiation extractions, but Morrish *et al.* (1981) felt pre-irradiation extractions did not lead to an increased incidence of ORN. Starke and Shannon (1977) reported one case of necrosis in 62 patients who had pre-irradiation extractions; there was an average healing time of 25.3 days. Beumer *et al.* (1979) suggest that 7–10 days healing is advisable, but where the radiation dose is going to exceed 6500 rads, 14–21 days should be allowed. Grant and Fletcher (1966) and Morrish *et al.* (1981) feel that post-irradiation extractions are more prone to cause ORN than pre-irradiation extractions, while MacComb (1962) felt that there was no difference. Later however, Daly *et al.* (1976) showed the incidence of ORN following post-irradiation extractions to be lower than that following pre-irradiation extractions. There is general agreement that antibiotic cover for post-irradiation extractions is advisable but opinion on extraction technique both pre- and post-irradiation is divided. Coffin (1964; 1983) emphasises a conservative approach with 'atraumatic' non-surgical extractions. Others feel that alveolectomy and removal of all sharp bone, with edge to edge closure of the soft tissues over the socket is preferable (Beumer *et al.*, 1972; Horiot *et al.*, 1981).

Materials and Methods

The population of the North West region of England is just over 4 million. All patients in the region come to one centre for radiotherapy. Between 1973–1978 this centre carried out a trial of prophylactic neck node irradiation. Patients presenting with new squamous cell carcinoma of the oral cavity who were otherwise fairly fit and clinically had no neck node involvement were included in the trial. 205

patients, half of whom received prophylactic neck node irradiation, were involved and this group formed the basis for the present study.

The study was carried out in two phases. First, all records were examined and basic information relating to age, sex, site of tumour and treatment regime was obtained for each patient. The records were often incomplete from a dental point of view and therefore no assessment of dental status was made at this stage. Any incident of ORN recorded in the notes was noted.

In the second phase of the study carried out in 1983, the surviving patients were invited to attend for examination. Patients who failed to attend or who lived in areas remote from the examination centres were sent questionnaires. In the questionnaires the patients were required to give simple 'yes' or 'no' answers to 10 questions.

Both the examinations and the questionnaires were designed to find the following information: the presence or absence of teeth at the time of radiotherapy, the history of denture wearing, the relationship of extractions to radiotherapy and the presence of signs or symptoms of necrosis at any time since radiotherapy. During all stages of the survey an attempt was made to divide ORN into major, moderate and minor groups. Minor ORN consisted of ulceration with exposed bone and a history of loss of bony spicules which healed spontaneously over a period of months. Moderate ORN consisted of exposed bone and small sequestra limited in nature and healing spontaneously or with conservative treatment within 6 months – one year. Major necrosis was large areas of exposed bone, with formation of large sequestra, possible fracture, and sinus formation. The condition was often rapidly progressive lasting in excess of one year and often requiring radical treatment.

Results

Part 1: Examination of Records

The records of 205 patients were examined. Six were excluded for various reasons including inadequate records, development of a second primary tumour and early surgical intervention. The age of the patients ranged from 25–74 years with an average age of 59.8 years. One hundred and thirty-six were male, giving a male to female sex ratio of 2.2:1. The tumour sites and stages are shown in Table II.

One hundred and thirty-seven patients received megavoltage therapy from a linear accelerator giving a tumour dose of 5500 rads over a 21 day period. Sixty-two patients received interstitial therapy with a radium needle implant. These were

Table II
Site and stage* of tumours

Site	T1	T2	T3	Total
Tongue	33	33	4	70 (35%)
Floor of Mouth	12	37	7	56 (28%)
Alveolus	5	21	6	32 (16%)
Buccal Mucosa	3	21	6	30 (15%)
Other	1	9	1	11 (6%)
Total	54 (27.1%)	121 (60.8%)	24 (12.1%)	199

*T1: 2cm or less; T2: 2–4 cm; T3: 4cm+

placed mainly in the tongue tumours and delivered a tumour dose of 5000–7000 rads over 7 days. Approximately half of the patients in each group received prophylactic neck node irradiation of 5000 rads.

The records of 36 patients (18.1%) showed evidence of ORN. Of these, 11 (5.5%) had minor ORN, 10 (5%) moderate ORN and 15 (7.5%) had major ORN. Six patients (3%) had undergone hemimandibulectomy following the development of ORN. The average age of the patients with ORN was 59.3 years and the male to female sex ratio was 1.8:1.

The incidence of ORN at different tumour sites is shown in Table III. The incidence of ORN in floor of mouth and alveolar tumours was significantly higher than in other sites ($\chi^2=11.337$, df=1, p=<0.001).

The incidence of ORN occurring with tumours at different stages is shown in Table IV. The difference between the groups was not significant ($\chi^2=1.5306$, df=2, p=<0.25). The relationship between the incidence of ORN and the treatment is shown in Table V. There was a highly significant difference in incidence of ORN between the megavoltage and implant therapy groups ($\chi^2=16.5015$, df=1,

Table III
The incidence of ORN at different tumour sites

Site	Total	ORN	%
Tongue	70	5	7.1
Floor of Mouth	56	17	30.3
Alveolus	32	8	25
Buccal Mucosa	30	5	16.7
Other	11	1	9.1
Total	199	36	

Table IV
Tumour stage and incidence of ORN

Stage	Total	ORN	%
T1	54	7	12.96
T2	121	25	20.66
T3	24	4	16.66
Total	199	36	

Table V
Treatment and the incidence of ORN

Treatment	Total	ORN	%
Megavoltage	70	17	24.3
Megavoltage + Neck XRT	67	18	26.9
Implant	30	1	3.3
Implant + Neck XRT	32	0	0
Total	199	36	

p=<0.001). Of those treated with megavoltage therapy, there was no significant difference in the incidence of ORN between those receiving prophylactic neck node irradiation and those not. Likewise, with patients receiving implants, prophylactic neck node irradiation did not affect the incidence of ORN. It was noted that in 6 of the 36 patients (16.6%) with ORN, necrosis occurred following surgery either at the original site or block neck dissection. The time of onset of necrosis varied from 2 months – 5 years following radiotherapy with an average of 19.4 months.

Part 2: Examination of patients and questionnaires

In 1983, 5–10 years after therapy 111 patients were dead, not necessarily as a result of their tumours. Of the 88 patients still alive, 72 were contacted and form the basis of the second group. Thirty five (48.6%) attended for review and 37 (51.4%) answered a questionnaire. These patients had an average age of 57.3 years at treatment. The male to female ratio was 1.2:1. The sites and stages of their tumours are shown in Table VI and it will be noted that there were more patients with tongue tumours and T1 stage tumours than in the original group.

Seventeen patients were found to have a history of ORN. Most of them (14) had been identified in the previous examination of the records but in addition there were three new cases. Of the total of 17 patients, 10 had active ORN at this time; in the other seven patients the episode had settled. Although the proportion of tumours at different sites had changed in this sub-group there was still a significantly higher incidence of ORN in floor of mouth and alveolar tumours than at other sites (χ^2=13.920, df=1, p=<0.001). In this group it was noted that there was an increase in the incidence of ORN with tumour size (T stage) which was significant at the 5% level (Yates χ^2=6.502, df=2, p=<0.05).

Of the 72 patients, 20 had some or all of their natural teeth at the time they received irradiation, six of these patients (30%), subsequently developed ORN. In the 52 edentulous patients ORN developed in 11 (21%). The difference is not significant (Yates χ^2=0.628, df=1, p=<0.5). Even if those undergoing pre-irradiation clearance of all teeth are not considered in the edentulous group there is still no significant difference (χ^2=1.493, df=1, p=<0.25). However, it must be noted that distribution of the tumours by site between the dentate and the edentulous groups was not equal (Table VII), and tumour site has already been shown to be a significant factor in the development of ORN. It also should be noted that all the dentate patients in this group with floor of mouth tumours developed ORN. Of the patients whose ORN was still persisting in 1983, 5–10 years after

Table VI
Group 2. Site and stage of tumours

Site	T1	T2	T3	Total	%
Tongue	21	16	—	37	51.4
Floor of Mouth	5	8	1	14	19.4
Alveolus	3	6	1	10	13.9
Buccal Mucosa	2	5	1	8	11.1
Other	—	3	—	3	4.2
Total	31	38	3	72	
%	(43%)	(52.8%)	(4.2%)		

Table VII
Group 2: The incidence of ORN at different tumour sites in dentate and edentulous patients

Site	Dental Status	Total	ORN
Tongue	Dentate	13	—
	Edentulous	24	2
Floor of Mouth	Dentate	5	5
	Edentulous	9	3
Alveolus	Dentate	2	1
	Edentulous	8	3
Buccal Mucosa	Dentate	—	—
	Edentulous	8	2
Other	Dentate	—	—
	Edentulous	3	1
Total	Dentate	20	6
	Edentulous	52	11

treatment, 70% had had a floor of mouth tumour. Teeth were directly implicated in the development of ORN in eight of the 17 cases of ORN in this group. In six cases extractions were responsible for initiating the ORN and in the other two cases marked dental sepsis appeared to be the problem. Five cases were the result of denture trauma particularly the wearing of dentures almost immediately after radiotherapy. Four cases appeared to have developed spontaneously, but in two of these it was possible that denture wearing had been contributory. In nine of the 17 cases ORN developed within the first year after treatment. The others occurred at various times up to 8 years post-irradiation.

In this survey, 36 patients were found to have developed ORN when the records were examined, and three new cases were found, giving a total of 39 patients who developed ORN from a group of 199 an incidence of 19.6%.

Discussion

The incidence of ORN found in this survey coincides with many other surveys (Table I). Coffin (1983) reports an exceptionally low incidence, but his criteria for diagnosis of ORN were those cases severe enough to warrant resection. Bedwinek *et al.* (1976) report an overall incidence of ORN of 14.2%, but only 2.8% required resection.

All patients in this survey who had megavoltage therapy received a dose of 5500 rads in 16 fractions over 3 weeks. Bedwinek *et al.* (1976) found spontaneous necrosis did not occur with doses of less than 6000 rads given over 6 weeks, and was uncommon at doses of 7000 rads in 7 weeks. Morrish *et al.* (1981), who reported an incidence of 22% had no cases of ORN at less than 6500 rads. In their patients, 5000–7500 rads were delivered at 180 rads per fraction, five fractions per week. In our patients the total dose received was at the lower end of the range at which ORN appears to become a problem; however the dose per fraction was radiobiologically considerably higher. Beumer *et al.* (1972), whose incidence of 3.6% was lower than many other reported series gave 5000–7000 rads over 6–8 weeks.

In this survey the incidence of ORN in patients receiving implants was very low. In other surveys implants have been reported as giving a high incidence of ORN. In this survey, however, in common with most other surveys, tumour sites close to

bone have been identified as giving rise to an increased incidence of ORN. In the cases reported here implants were used mainly in tongue lesions well away from bone.

In some surveys where tumour size is reported as having an effect on incidence of ORN, larger tumours had been treated with higher doses of radiation (MacComb, 1962; Grant & Fletcher, 1966). Rankow and Weissman (1971) report size of tumour having an effect, and 75% of their cases received 5000 rads only. In this survey, in the whole group there was an increase in the incidence of ORN in T2 and T3 compared with T1 tumours, but this was not statistically significant. In the subgroup, however, the difference was significant. In this survey there were no very large tumours and the results are equivocal. It is felt, however, that even using the same radiation dose to treat all tumours there is an increased risk of mucosal breakdown occurring in the centre of large tumours which may lead to overt bone necrosis.

In this survey there was a fairly high incidence of ORN in edentulous patients. Most of the necrosis in these patients was in the minor and moderate groups. It was felt that much of this was due to patients wearing what were often very old and ill fitting prostheses immediately the mucosal reaction had settled. Beumer *et al.* (1979) suggest patients wait 6 months – one year if they were edentulous prior to the onset of the tumour, before new dentures are inserted. They advise a longer period for those who have undergone extractions associated with radiotherapy. The incidence of ORN was increased for patients who had extractions both before and after radiotherapy.

Conclusion

The incidence of ORN in the North West region of England has been shown to be 19.6%, which is within the range reported by others. The patient most at risk was one who had a tumour in the floor of mouth or on the alveolus which was to be treated by external beam therapy and who also required extraction of teeth. There appears to be little to choose between pre- and post-irradiation extraction. ORN is likely to appear within the first 2 years after treatment but is not generally apparent in the immediate post-treatment phase unless related to the extraction of teeth or a surgical procedure.

Acknowledgements

I would like to thank Dr R. Pointon, Director of Radiotherapy at the Christie Hospital, Manchester, for allowing me to review these patients, and Mr W. Simpson, Consultant Oral Surgeon for his help at all stages of this study.

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